

ActInf Textbook Group ~ Cohort 6 ~ Session 3 (Chapter 1, Part 2) ~ 2/12/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 3 (Chapter 1, Part 2) ~ 2/12/2024 | 59:10

all right welcome

everyone it's cohort 6 and our second

discussion of chapter one so Andrew and

Ali thanks for facilitating

just take it from here wherever you want

to pick

up Andrew would you mind taking the lead

on this one

sure yeah so I know um yeah absolutely

so welcome cohort 6 this is week two

that we're spending on chapter one of

the

textbook um I'm not sure what was

discussed at the last meeting but I

thought it might be worthwhile for some

of the folks who are unable to attend

the morning meetings to maybe just

quickly move through um the sections of

textbook

um just to quickly get a bit of a recap

or

overview

um yeah the overview we begin with a

nice brief

introduction um describing what's going

to be happening in this chapter we

already get off the bat that active

inference is a normative perspective

it's starting from first

principles um move on to section

1.2 get some quick explanations

regarding the relationships between

things like actions and sensory

observations specifically that living

organisms can only maintain their bodily

Integrity by exerting adaptive control over the action perception Loop this means acting to solicit sensory observations that either correspond to desired outcomes or goals such as the sensations that accompany secure nutrients and shelter for simple organisms or friends in jobs for more complex ones or help in making sense of world um this is carried out through there's a rather recursive nature to this cycle of action uh perception Loops um also whenever we move on to looking at more complex organisms there's the assumption that organisms can increasingly use like more complex adaptive strategies to adjust to their environment um section 1.3 still more on um first principles get an introduction to this uh this kind of I don't want to call it ad hoc because I think these conversations have been had before but it's a a little light-hearted I suppose this meets versus scruffy's uh division that's main right and so um scruffies being like these are folks who will approach these kinds of problems but maybe from within their particular field or subfield um they don't want to um you know forego any kind of kind of idiocy idiosyncrasies or specificities to their object of study um so maybe you know not

so quick to jump to some kind of broader unifying framework um to analyze things whenever you know you might let go of some of the specific the problem you're looking at um you anthropologists will have disagreements with sociologists and so on about what what defines like a society or or a culture or otherwise um nonetheless meats are those who do seek unification Beyond heterogeneity of brain and mind phenomena and that is the perspective that active in for instincts it does in fact attempt a kind of unified framework at looking at diverse uh scenarios and organisms ranging from human beings and entire cultures uh who are operating at different time scales of the way at fast time scales of like what they're immediately observing and perceiving um all the way up to slower time scales of like planning learning um organizing with a group development of Institutions and so on um section 1.4 gives us a good overview kind of structure of the book

um

1.4.1 active inference in theory we get more about surprise minimization um we're also introduced to variational free energy in the next paragraph there um they keep it short in describing why we might look at variational free energy as opposed to surprise which seems like a much more intuitive concept um for one to be surprised at unexpected uh phenomena or

observations or incoming um sensory signals um but the the to put it quickly variational free energy is basically an upper bound on surprisal that is done because computationally um we find this issue of intractability in computation when whenever you break down the math you find like oh in order to actually precisely compute surprisal you would need something like an entire history of an organism's like memory every observation it's come across every potential in state it would estimate and you can imagine that the the the the mathematics to that the kind of formula that would be involved in that would be um just untenable from a computational perspective and you can imagine the same thing for an actual human brain um and making snap judgments based on its entire history of what it's learned and so on so variational free energy is is very important in order to overcome this kind of intractability problem everything is somewhat approximate with the goal being to minimize free energy as much as you can such as surprises also minimized as much as possible there will be much more on that in the textbook we're only in chapter one um so stepping forward from that um we do have this nice figure 1.2 which describes um how to reach active inference from two different routes uh the low road and the high road we can say that the low road is kind of the the how of active inference how an

organism maintains its own Integrity um
just before we started this meeting
there were certain uh mentions of autop
pois and so on like this is the how and
and so it it relies heavily upon basis
theorem uh we work up to generative
models and using bases theorem in order
to find relationships between different
variables within the model um and so
that's kind of the mechanics it's the
how but from The High Road alternatively
we get this is where things become a
little bit more of aose uh which is that
this is what and why organisms are doing
what they're doing as far as their
behavior goes and attempting to explain
like what is happening here um they're
minimizing free energy so we begin with
the free energy principle right so we so
then we can start understand living
organisms or other kinds of systems that
we attempt to model using these kinds of
tools on the one hand they're trying to
minimize free energy on the other hand
how are they doing it they're doing it
through something like Basi and
statistics mechanics
um we'll get much more on on both of
those roads to active inference in
chapters 2 and three Each of which are
dedicated to one route uh one route
each following that learn that yourself
when we look at the structure of the
book as it's described from there um to
keep it a little bit shorter um I'll
just say the the first half of the book
is um predominantly dedicated to a lot
of the more theore iCal aspects of

active inference um there will be
explanation of other aspects such as the
the mathematics involved although you'll
find much more on the details of that
math in the appendix um the active
inference Institute also um if you view
the Koda you'll find like references to
the maths um you'll see that there's an
ongoing project right now and it's being
streamed regarding the mathematics of of
active inference so there are plenty of
resources outside of this textbook um
for those who are more interested or or
feel like they're not getting a
sufficient amount of explanation from
um and then we'll also later in this
first part of the book we'll also get
references to uh the kind of the more
neurobiological aspects of the theory of
course it was um largely initially
devised by Carl friston as a
neuroscientist um
a neurobiologist and so the all of this
is heavily based within preceding work
and looking at how to understand the
mind and the Brain
um so with that
said I think beyond that we'll we you
know key words include active inference
is of process Theory we're looking at
different kinds of models discret and
continuous
models we're getting references to yeah
just a variety of things that this is
connected to Hybrid models um I think
that might be fair enough we're trying
to introduce the the chapter we're
already nearly a quarter of the way

through so

um so I'd like to ask if there's any are
there any questions that anyone has
about the textbook or anything that
they've kind of been chewing on for some
time you'd like to

discuss um so can surprise be seen as
the difference between expected
observations and actual

observations yeah I'm sorry I missed
just the first couple words of what you
said could you repeat that for me please

yeah um so so I was asking um can
surprise be seen as the difference
between expected observations and actual

observations that's right so so active
inference kind of relies upon this um

idea that you can think about it as um

like a like a set point for example um

in in control theory or or looking at a

thermostat right so you set the

thermostat to a particular temperature

um there's a signal that's being read

about the temperature in the room um

here's what you want the temperature to
be

here is what it actually is um and then

the idea is you'd want a heating or

cooling system that could then take

action here to bring the the temperature

that it's reading in line with the

expected set point temperature that you

want um and you can quickly see like

yeah you're you're minimizing the

difference between expected and actual

it's a kind of classic machine learning

persp perspective or or or otherwise um

yeah that you'd be effectively

minimizing

surprise um in that

sense but mathema Ma sorry

mathematically it's easier to minimize

variational freey than it is to try and

minimize the area directly right that's

right y the textbook will get much more

into that but it's it's that itself is

also very much based upon mathematics

it's not a arbitrary decisions so but it

it has to do with Jensen's

inequality um the idea that um basically

a particular quantity is always

necessarily greater than or equal to

another one given certain conditions and

so that's how we end up there

um thank

you I ask something uh sorry can you

hear me

yes yes uh i u a lot of things hinge on

the generative model that an agent or an

organism has and I I often wonder it

like and and yeah when I read further in

the book then I see that the generative

model is considered given somehow you

you you seem to you you seem it's

considered to that you have access to it

uh so then I wonder where does that

model come from who chooses what model

is there uh the parameters seem to be

updated through message passing and that

kind of uh stuff but uh uh the model

itself um it's the the P and not the Q

but where where does

it yeah where does it come from or or

how should one think about

that

before I give an answer I was just

wondering if anyone else um wants to
chime in on that thought because it is a
big
question it's um I guess uh does the
free under oh I'm sorry people are
raising their hands uh let them go
first yeah
Ali Anna did you have a
question oh yeah but Ali was is gonna go
first he was
first uh okay so what I wanted to
comment
was a bit unrelated to this current
question so Eric would you please uh go
on about what you were talking about
before
thanks oh I I was just going to say
um I think is his like how the model is
updated is is that
um is that really addressed directly by
the free energy principle or is it kind
of like mysterious right
because um it seems like uh you kept on
you keep on getting this phrasing from
uh friston like this is how a an
organism has to behave if you know um so
uh um I don't know coming from a
neuroscience perspective I'm not sure we
quite understand how it
works but we could see that the brain
seems to be doing it
something like
this I I don't know if that's a good
answer
but yeah know that that makes a lot of
sense I mean it's so to I mean come at
to come at any kind of and this is just
me speaking but to come at any kind of

phenomena I presume you have to you know
you start with some kind of framework
where you attempt to like kind of
structure things and then and then as
far as living organisms and the behavior
goes yeah it's I mean all of this is
sort of um you know the map is not the
the territory we're just attempting to
understand the organisms right and so
active inference is like attempting to
provide that so the just to comment on
how you phrased it as like it's as if uh
it is undertaking free energy
minimization I think that's a just fine
way of saying it um so as far as yeah
defining a generative model I
suppose it depends on what kind of
perspective you want to take so there
are many people at the Institute who are
more focused on the modeling and and the
programming right and so they want to
construct like a model themselves U
maybe it's to simulate something like an
agent it's an environment um in that
case yeah you're kind of you would want
to rely upon things like domain
knowledge in in conjunction with with a
lot of the principles of active
inference and basing mechanics which as
we mentioned earlier is kind of the low
road to active inference um that's
already your starting point you you
already assume that the agent is
attempting to minimize free energy and
it's doing it through a variety of
principles all related back to basis
theorem um so that that's kind of the
the top- down normative framework that's

being used here

and then from there it's sort of up to you as far as how you want to continue constructing that model right so if if if there's certain actions that are available to the model or not that it can take um what are the different kinds of states it it already expects to see what are the the priors this this is the kind of language that we'll get more deeply into is the the textbook you know as we continue reading through the chapters

um from another perspective if you're not modeling but rather just kind of going through the theory here I mean it's there can be bigger philosophical questions about um the structure of an agent why it is the way it is um I think that there are many people who study active inference who kind of follow the line of of something along the lines of natural selection or perhaps more nuanced um versions of that theory uh you know since its initial conception and so I don't want to say too much here but has that at all been helpful in in trying to answer the initial question

yours yeah I I see that this is uh this is not not an easy question and not uh but I I understand that there is for a modeler there is a lot of room for uh you say creative filling in of the structure of the model at least it seems uh while um yeah if I get you right then it's not

something that is necessarily derived
from uh a
normative point of view or
something yeah I mean of of course like
with any kind of scholarly pursued I
mean whenever there's already work
that's been done whenever there's
already secondary literature to draw
from and so on of course folks will
build upon one another's models and test
things out and oftentimes you'll
construct one model and then eventually
realize like it's Its Behavior despite
the fact that it's minimizing free
energy the way you might have set up the
variables in the model like the behavior
at elicits doesn't actually match any
kind of real world phenomena that you're
actually you might be attempting to
simulate for example um in that case
it's it can be a trial and error process
um it can involve discovering something
that you didn't expect to find when
you realize oh there's a missing
variable here um yeah those are many of
general uh trials and errors of of model
construction I suppose um yeah
thank

you um Anna I did want to come back
around to you yeah
um okay so actually my

[Music]

question um is relevant to the previous
question I think um so on page six of
the book um there's you know this
sentence that I'm going to read it says
active inference is a normative
framework to characterize Bas optimal

behavior and cognition in living organisms and so I was wondering if you could um give us a an overview or some um ideas to start thinking around the you know what what does a basian model optimize and um so because that's the basian optimization is a tool that allows us to think about the world and so I'm curious about things things like what what utilities are uh optimized in basian models uh is that are the utilities um specific to the phenomena that you're working with um and and if you could connect and discuss you know the utility to the probability functions that are used in basian stats statistics so again this time I do want to open the floor if anyone has any any thoughts on on this or otherwise um to maybe see if we can get into this um this question that an's posed go foron oh well I I had my hand raised from to ask an independent thing but I guess like in terms of um I can try to take a stab at this one um so I would I would phrase it as beian models don't on their own optimize for anything but the model itself States the relationship between variable that is assumed and under those assumptions one can in light of new evidence um Express a posterior

distribution which describes one's beliefs and I think classically that's like fully separate from how we think about utility uh utility functions and sort of given my posterior distribution about the world I can then choose to act depending on what my my preferences are

can you can you um add to your example like

uh a you some you know what what is the currency that's being optimized it just I understand that the modeler decides what currency is being optimized So when you say when you're talking about beliefs updating beliefs and so on what are we optimizing

there you know so how do we translate uh the you know the natur the natural language um updating beliefs what does that look like in a in a in a model and

can so one one way that I would anyone feel pry derail to derail me on this uh one way that I think about um ban updating and and I think like the currency that that's at play here is uh units of surprise so bits or Nats right and I think like the the the textbook doesn't really talk about this I feel like other sources are better about like grounding bits or kns as like a unit for surprise but like if you were to have a lot of observations from a distribution and you want to communicate those to some other party efficiently um

literally how many bits do you need per
per observation um that's the kind of
thing that uh is reduced as your model
of the source of the data is improved so
if you're if you're constantly surprised
then you're going to take more bits per
observation to communicate what you're
observing and if your model is pretty
well explaining what you're seeing then
you'll need many fewer bits per
observation to to report or communic
um and I guess maybe communication comes
like loaded with some expectations of
like between agents but I think also
that can be between components within an
agent in in some cases but yeah that
bits or kns are like actually an
expression of uh perplexity surprise um
on average I suppose um that can be uh
reduced by updating
appropriately thank

you

okay and I I do want to add to that to
um I think it was well said Erin in
pointing to yeah it's it's frequently
understated like what the units of
measurement here are whenever we're
looking at something like surprisal um
that said if we want to look at it in
terms of active inference I mean some
some things that you're going to see are
are commonalities throughout the
textbook is we're going to be looking at
particular variables and so um we're
going to be looking at priors which can
be viewed as something like prior
predictions what you think is going to
happen um or or is happening um I.E your

hidden state that you're estimating and then you'll have sensory observations right which is new information that's coming in um this is where we get into things like predictive coding or processing at least uh where there is a continuous process of um comparison between what information you're getting versus what you thought would happen this kind of hearkens back a little bit to the general idea of you know here's the expected observations here's the actual ones like the thermostat examples and it's through kind of starting with bases theorem as a sort of foundation like that's the how of of reducing surprise right so um given some new information exactly I mean I don't want to rehash what Aaron said because I think you put it much more eloquently than what I'm prepared to but um yeah it's to

um to minimize surprise via uh taking in new information updating what you thought you knew before suddenly your predictions perhaps come more into line with what you see out in the environment it's much more complex than that depending on the situation but um that's an attempt at getting closer to describing you know here's what we're doing whatever War minimizing surprise um Erin I do uh you've had your hand up for some time so I'd like to allow you to speak on on your original question yeah okay so one thing that I struggle to get my head around here is the generative model um is supposed to

generate like all of your sensory data
or is like supposed to like go all the
way out to like you know the the margins
of the mark blanket or whatever the
things that are coming in from the
exterior World um and how plausible is
that really like there's lots of cases
where you can see a complicated
situation and not be able to predict
most of it um and a lot of the things
that the details that may be surprising
possibly aren't going to be relevant to
any action uh you know you look at um a
whole bunch of birds flocking together
you don't know where any particular one
of them is but it kind of doesn't matter
uh if you look at a TV screen with a
bunch of static you can't predict what
color any pixel is going to be in the
next millisecond but you can predict like
derived features about a scenario um and
so I'm wondering sort of there should be
some intermediate layer uh that is more
relevant for our generative model to
populate than like the actual sense data
coming

in
does anyone have any thoughts on
that uh actually this is uh directly
relevant to John Verve's relevance
realization uh because um in the in that
framework uh it's
not it's not possible for an organism to
encode every single bit of information
about its
environment uh I mean with uh sufficient
Precision so uh it needs to only encode
uh only the information uh which is

relevant uh relevant uh to the uh
situation uh which it's dealing uh with
with currently so um one way to do that
is to uh is to encode those information
probabilistically instead of
deterministically because uh in
uh our traditional algorithmic
formulation of cognitivism for example
uh we deal with with well-defined
situation in which the parameters are
pretty much clearcut clearcut and we can
encode them uh with u arbitrary
Precision if you want but uh in one of
the selling points of predictive
processing Frameworks is that uh it's
not necessary
to uh to encode those information with
any uh kind of precision at all you can
even start from uh I mean in principle
you can start from a from an arbitrary
Precision from an arbitrary distribution
and then uh and then move your way uh
in in a in an optimization scenario uh
so for in for instance you can do a
gradient descent on the VF landscape and
reach uh the extrema or the minimum uh
of the variation of free energy and uh
that's uh one of the one of the
fundamental differences between uh
active inference or predictive uh
processing
Frameworks uh from the other traditional
uh cognitivist uh Frameworks
because uh as this recent uh paper
naturalized in relevance realization
shows
uh in real world real world
situations cognition might be

fundamentally incomp uncomputable so but
um we can uh I mean get around uh this
obstacle by uh somehow uh encoding uh
that generative model
probabilistically uh and only encode the
information that are relevant uh to us
yeah I think there there have been
arguments that have been made
especially just a brief tangent that
that you know living organism living
organisms who who necessarily like have
um kind of metabolic um rules to to keep
in place and you know home homeostasis
to maintain tendency to we we can view
this as one potential explanation for
something like only paying attention to
what is surprising and meanwhile trying
to kind of put on the back end all the
processes involved in um you know
determining like what is the new
information here and what is not the new
information do we need to know the
position of every single seagull in the
flock as it moves by to be able to
comprehend that this is a flock of
sequels right um it you know that I I
think it just it kind of hits on the
aspect of of active inference but it
doesn't even necessarily have to be
active inference like the importance of
something like a having a variational
dynamic that allows for things like
dimensionality reduction and increase
from there compressing information and
um just kind of keeping things a little
bit more
efficient um in any case um I would like
to give others a chance to ask questions

uh

how apologies if

I um yeah thanks so that um yes so I

guess you know I guess coming from

machine learning as well I I try to I

guess compartmentalize my my learning

here and um you does active influence in

influence news um new networks gradient

descent um and you like back

publication or is it like a just you

know

different I'll give a quick answer to

this there's an

essential analytical form that is just

matrix multiplication so what we see in

the textbook is just matrix

multiplication how however a lot of the

modern uh examples have used neural

networks as kind of like plug-and play

for different things like instead of a

massive a matrix you can have a neural

network and there's also deeper

theoretical Connections in live stream

51 with tuya isura this discusses the

monotonic relationship between the

variational free energy and the neural

network uh loss function and then

live stream 52 or like anything with

Lance D Costa is super informative about

how accelerated optimization methods are

used with variational free energy some

of which are um generally accessible to

any variational method others are

especially enabled by act in for

FP interesting so it can use new

networks but it it doesn't need to

exactly

oh

okay

um oh I had a question but yeah I've lost

it um yeah thanks for the answer by the way uh

Thomas yeah I mean just kind I want to build on some of the questions and looking for examples on that generative model point I remember the the frisen article on aesthetic experience uh Order and Chaos there's a note in it about the nature of absences so there's like visuals basically of a square with a piece cut off and there's a comment in the caption they talked about absences are meaningful as obstacles or prediction error so I think that's really interesting because that's pointing out that a kind of absence um it's almost like where's the grounding like where's the ground of mod come from it seems to be and that relates to the other parts of that essay where he talks about this kind of epistemic promise that our we're using we engage in AR the experiences because there's been a promise that we're going to get some kind of payoff of epistemic basically Clarity or resolution of some kind um there's a lot more Nuance in that but just I want to point that just to kind of build on this point where is this Jen of model ground bottom up thing coming from absences or lacks basically that are pointing out that that's a kind of course basically either there's an objective you're trying to achieve that's being hindered so there's an obstacle in that mode or

there's a basically a difference between
the the prediction model in your head
versus the reality of interfacing with
it um I think that's interest that's
kind of that was an interesting just
building point on where potentially a
generative model comes from um also just
other idea like where I'm thinking in my
head I was looking reading this piece
and where this comes from concretely as
an example it's like you got a gallon of
milk in the fridge you think it's full
you're not paying attention so you
recruit way too much energy boom and you
pull you You Yank it out with way more
Force than expected
uh that difference basically is a
surprise I mean that's why I'm thinking
as a concrete example of um so in that
case what's the currency being
manipulated years is calories basically
to bur basically to recruit the energy
required for the muscles to contract and
move on that example those things I
wanted to
share yeah no good good points than for
sharing those I mean yeah again with the
you know expecting a much
you know jug of of milk is another very
quick intuitive example of you know
here's the difference between what you
predict is is currently going on the
hidden state of the world like I have a
full jug of milk um what actually
happens there's much less uh milk in
there you know you end up like yanking
it out way too quickly there's excessive
metabolic like expenditure there um

those are all ways of for sure thinking about it and and you know uh at some point in the textbook we'll see explanations of things like uh Arc reflexes um and proprioception where uh it'll describe things like like for example you can move on from that example you gave to imagining like the arm will adjust right like well we hope it does anyway like the arm will not just immediately fling out it will like maybe halfway through a quarter of the way through the movement start to pick up like oh this is actually not heavy enough to justify this amount of force there'll be some kind of change in the action sequence right like suddenly you're not hopefully you're no longer G to fling it sorry it's kind of a silly example isn't it but um but you see the point it's like error correction right so so there's what you expect there's what's you're actually observing and then there's an um action can then be used to try and correct for that error um which ultimately uh presumably minimizes free energy along the way

Eric um yes I was just gonna also add um to the question raised earlier that maybe part of the answer to the question about whether um the model really has to um model every Century detail is just that there's there 's actually many models that are working together um in in in an organic system so um they don't all have to model everything um and uh like if I

understand uh B's take um I think he believes that like the um task task centered Network and the default mode Network which kind of take turns right when you're um when you're doing various activities that you know the um the uh default default mode network is basically aspectual using different things in the world and um like offering them up to the um task Focus Network which is selecting some of them um and that they're constantly mutually modeling each other so that the organism can move through a salience landscape but um you know um we tend to reduce organisms you know being conscious creatures we tend to think of the model as oh it's me the conscious me right but there's all these um my understanding is that there's all these um generative models right working beneath my Consciousness um allowing some things to appear in others you know um so maybe that's part of um how it works efficiently because there's many different models hopefully that's coherent yeah know it makes a lot of sense to me and I I think um you know this where we start to see a little bit more relationship with um yeah as you described raiki theories as well as you know if anyone's familiar with like Global workspace Theory basically this the idea that you can have a kind of uh like multiple systems within a single system or nested systems and um we're

already in chapter one briefly introduced to the idea of a Markov blanket but um as the textbook continues there'll be some more regarding Markov blankets and what what separates statist in this sense stat also statistically separates a system from another system and then also recognizing that there can be like nested systems within systems or systems who appear to be working cooperatively together and so in the context of like a human being um we only need a like a you know what's a quick example someone who uh is a cigarette smoker or something right like someone who on the one hand like and decides to quit that's sorry I'll keep it more brief than that um someone who decides they want to quit like top level hierarchical like higher order decision I want to quit I want to be done with this um this is good like that's what I want to do that's a higher order decision on the other hand uh you know give them you know roughly 15 minutes and they start experiencing a craving and as that time continues we have you know withdrawal symptoms and so on um suddenly the person will become potentially more ambivalent um there are different systems in the body that are expecting that nicotine uh to to come in right um and so you can have like multiple systems within a single human being or a single system that are potentially in conflict with one another cooperating with one another um it's really about what is the

phenomena you're interested in whenever
it comes to active inference and
applying it as a framework what is the
what is the system of for you and and
you have to decide like what defines
those boundaries like what Behavior am I
expecting to see what system am I trying
to to um simulate

here what potential nested systems are
within that broader system um so at risk
of over speaking I'll I'll uh give the
floor back but uh no I I think those are
all great questions and uh things to
think about

Ali yeah I just wanted to add uh to what
Thomas just mentioned about um I mean uh
the the absence of information uh
because that's exactly uh the premise of
U Terence Deacon's incomplete nature
because uh in his magnumopus he develops
theory of absenti-ism and he tries to
somehow uh derive kind Tio Dynamics
based exactly on those uh what he calls
appsential so uh in one of our works
with uh ma and I U which is INF
interaction inference uh we uh we
initially developed an account of the
interaction between the organism and the
environment based on possibility spaces
so uh very briefly any system with a
given State space uh would would have a
form of the possibility space based on
uh the definition or the collection of
uh its Tendencies and capacities and uh
and the structure of this possibility
space can be described uh or defined as
a multiplicity as a delusion uh
multiplicity so by multiplicity uh what

we mean is a nested set of vector Fields
uh which are related uh to each other
hierarchically by symmetry breaking
bifurcations uh and uh which together
they constitute a kind of a collection
of distributions of
attractors um uh which Define each of
the uh their success their subsequent or
embedded levels so in in other words uh
the idea of essentialism is already um
embedded in these uh spaces of
possibility ities or the multiplicities
because those are exactly what
constitute the structure of uh the state
of the space of possibility as uh as the
attractors of the multiplicities so uh
in one of our recent papers we we're
trying to develop uh this connection um
a bit more explicitly and uh yes uh I
agree that um those appsential are uh
quite essential for uh any kind of
representational framework uh or
generative
models so I have a question Ali about
absent
ISM and sure uh yeah
so how can I ask this
um so in Bas in statistics we are
using probability
distributions
to get
at understanding something about
reality um can
you define
entals in the context of probability
Theory or is it completely
unrelated okay so uh first of all as um
as a side note uh Terrence Deacon

actually uh doesn't use the term abs
essentialism in his book and uh he even
rejects uh this word explicitly but um
in any case in all of the literature
related to incomplete nature they've
used the word absenti ISM so uh it's
pretty much um widely accepted so uh by
absenti ISM he means something that's
not uh physically realized yet or in
other words uh as would call
it it's not actual but a virtual
possibility of uh the system so uh for
instance as a very simple example you
can think of the capacity of a knife uh
to cut to kill or I don't know
to to poke something so uh so yeah all
of these are kind of capacities of the
the of the knife uh which are not always
actualized they reside as uh as its
virtualities uh in its space of
possibilities right and uh the knife may
or may not ever actualize these
virtualities but it uh it doesn't make
it any less real so by appsential
absenti Concepts um or as another Rel
term of Deacon um um intentional
Concepts he means something similar uh
kind of possibility or potentiality uh
which is not actualized or realized yet
uh but uh it is real and it constitute
uh I mean
the uh the D the internal dynamics of
the system uh as um as its potential as
its capacities and
Tendencies so uh I'm not sure if we can
characterize uh absenti ISM in terms of
probabilities because uh although it it
can be

said that some of those
virtualities uh are more probable than
others to be actualized uh but uh but it
needs um a much more rigorous uh work on
that uh to uh to derive these kinds of
probabilistic landscape over those
multiplicities but in any case that
multiplicity exists as a kind of reality
of the system or the agent uh again
whether uh I mean regardless of U
whether it actualizes those
possibilities or virtualities or
not
it would would it be incorrect to say
that there are states that are virtually
or have a
potential that they
exist in some virtual representation
somewhere and and that those states are
not uh accessible to us for whatever
reason
for to The Observer at some point in
time but yes to me to me that that
translates let's see I'm trying to see
if I this is making any sense to me that
translates
into in order for me to even think about
that thing that's absent I The Observer
need to have some sense of what what
could possibly what could
possibly be
absent so I as the Observer have to also
have a virtual version of the thing
that's
absent and exactly and would and would
that thing that's
absent not have a probability
distribution so you see uh these

appresential kind of reside uh
in the space of possibility for this the
space of possibility of the system right
uh but uh in order to actualize those
possibilities of course they can be
prioritized based on the probabilities
of U I mean those absent to be
actualized as I mentioned earlier some
of those
virtualities uh can occur with a higher
frequency than others right uh
but this is this is a very subtle point
and active inference
uh which is to say that uh you see in
recent years I mean uh beginning from
2022 uh there's this uh attempt to
somehow um blur the line between uh The
Observer and The observed by showing a
complete Symmetry and Duality between Fe
and cmap uh which is the constraint
maximum entropy principle right so in
other words uh what the the recent
formalism of active inference or rather
FP shows is this complete Symmetry and
Duality between uh these two perceptions
these two I mean distinctions between
observed and the Observer so uh every
time we uh draw a mark of blanket a
statistical boundary between a system or
between the Observer and The observed it
can basically be treated as a kind of
arbitrary boundary we can uh we can
switch that observed uh and observer
perspective um and uh the formalism uh
will help perfectly so uh um yes there
is this Duality and perfect symmetry
between Observer and observed which is
shown uh formally in recent work by

Dalton Sac divided EI and others and especially from the Viewpoint of category theoretical perspective it can be um shown much more eloquently uh through those um category theoretical formulation as well such as the work of um uh Toby SMI and others thank you appreciate

it

um unfortunately I see we're just about out of time here it's been we' run through the the whole hour and for a I will say for a discussion on chapter one of a textbook this got rather deep uh very interesting stuff though very happy to hear everyone share their their thoughts and uh

I would you know of course everyone here is coming from their own distinct perspective and and interests so I would say you know if you are coming from a certain angle certainly don't hesitate to to ask questions I think that's what'll well what will kind of pull you into to active inference a little bit more quickly in a way that you know is a little bit more fluent um for a lot of people because it can be a lot of material at one time but um thanks Ali for for Co facilitating and and your very U your very thorough responses I always end up learning something new about uh active inferences history and and sub theories that makes me want to read more um and uh thanks everyone else uh for attending uh per usual there will be another meeting next week uh we'll be going over chapter two which is where

we'll get into the low road to active
inference a lot more on basian
statistics and mechanics
um we'll get a maybe a clearer view of
what optimality means or optimal uh
behavior and inference um we'll also get
a couple perspectives on things like uh
different time scales such as inference
at a very quick Moment by moment scale
all the way up to uh planning and agents
who actually have the capacity for
planning their next move or or or
constructing you know goals that will
take a series of steps to carry out not
just taking a breath in the moment
because you need it um Daniel do you
have anything to to add or
announcements no great all right thank
you everyone farewell thanks every
thanks everyone bye